

Challenge 1

Build a ETL

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Constitution

Mision

A company need to applies a 10% discount to the first 500 online orders placed through the platform. Only the first order per customer per day counts toward the threshold.

Tech Stack

- Data extraction via kaggle hub**
- Jupyter Notebook built with Python (pandas, NumPy, SQLAlchemy)**
- PostgreSQL database hosted on Neon serverless, connected via psycopg2**

Roadmap

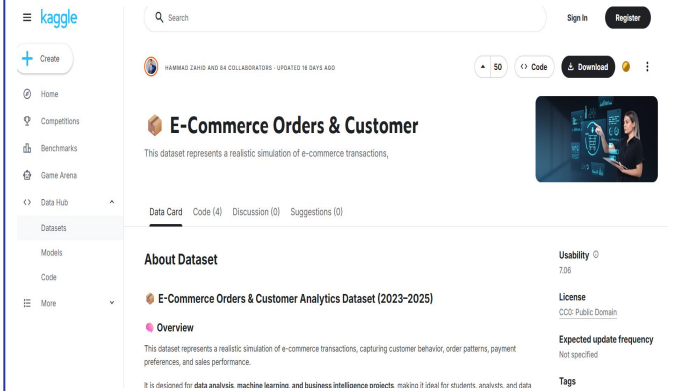
- Extract the dataset from the Kaggle platform**
- Build a Jupyter notebook ETL pipeline (Extract → Transform → Load)**
- Apply discount business logic and load data into a Neon PostgreSQL database using a medallion architecture (Bronze / Silver / Gold)**

Feature Phase

Phase 1

Task 1

Download the e-commerce dataset from Kaggle using kaggle hub and cache it locally



The screenshot shows the Kaggle interface for the 'E-Commerce Orders & Customer' dataset. The left sidebar contains navigation links: Home, Competitions, Benchmarks, Game Arena, Data Hub, and Datasets. The main content area displays the dataset title, a description, and a 'Download' button. Below the title, there are tabs for 'Data Card', 'Code (4)', 'Discussion (0)', and 'Suggestions (0)'. The 'About Dataset' section provides details about the dataset's origin and purpose. The 'Usability' section shows a rating of 7.06. The 'License' section indicates 'CC0: Public Domain'. The 'Expected update frequency' is listed as 'Not specified'. The 'Tags' section is also visible.

Task 2

Rename columns, cast types, deduplicate by (date, customer), and apply the 10% discount logic for the first 500 online orders.

Feature Phase

Phase 1

Task 3

Write data through a medallion architecture:

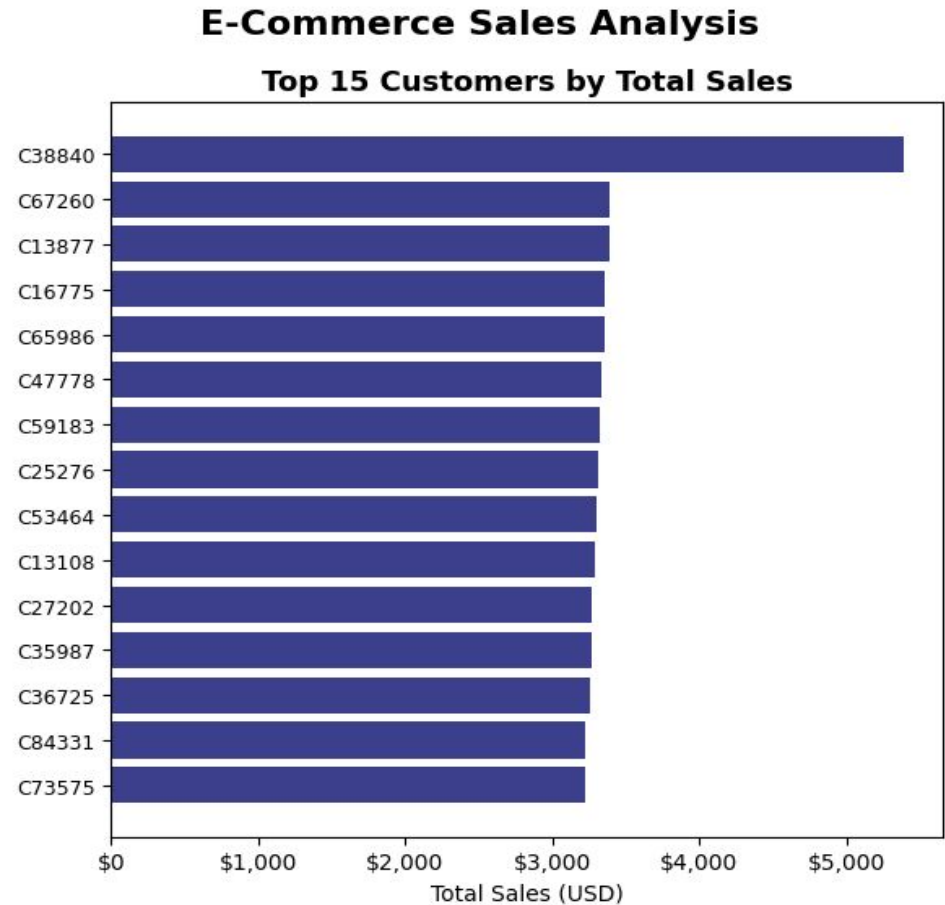
- **Bronze:** raw deduplicated data (raw_ventas)
- **Silver:** enriched with discount columns (silver_ventas)
- **Gold:** star schema (dim_cliente, dim_producto, fact_ventas)

Task 4

Query the star schema on Neon for business metrics: sales by customer, by product, and by day.

Total sales by customer

Ranks customers by total amount spent (final_total) across all their orders.

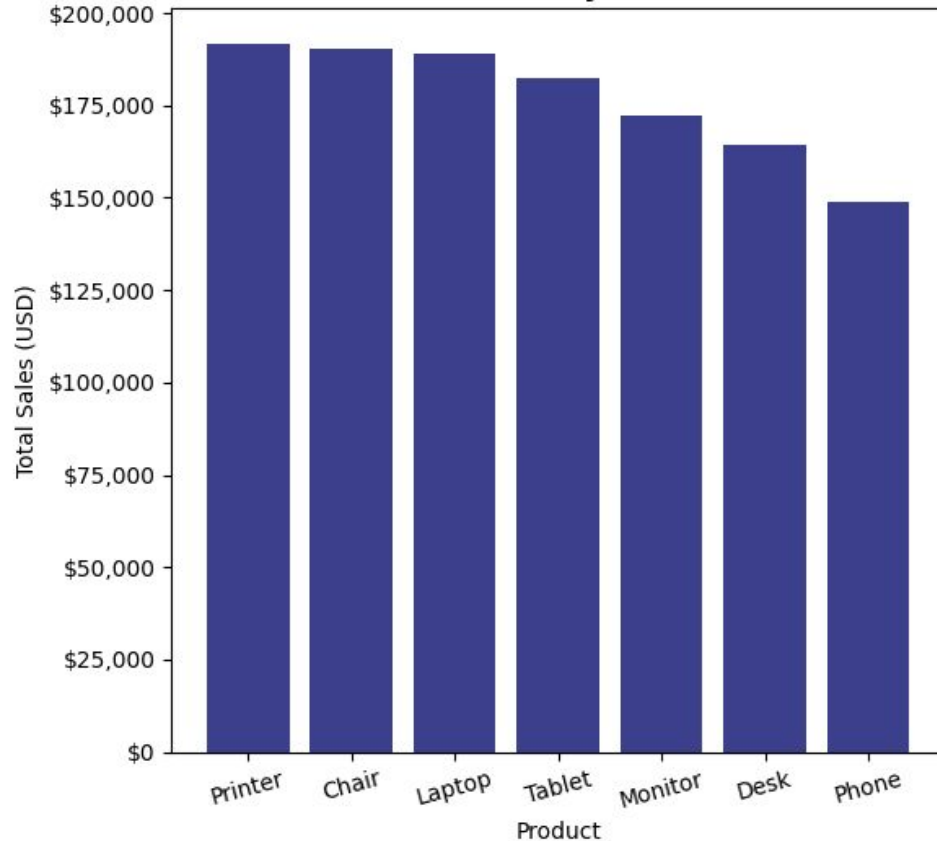


Total sales by product

Ranks products by total revenue generated

E-Commerce Sales Analysis

Total Sales by Product

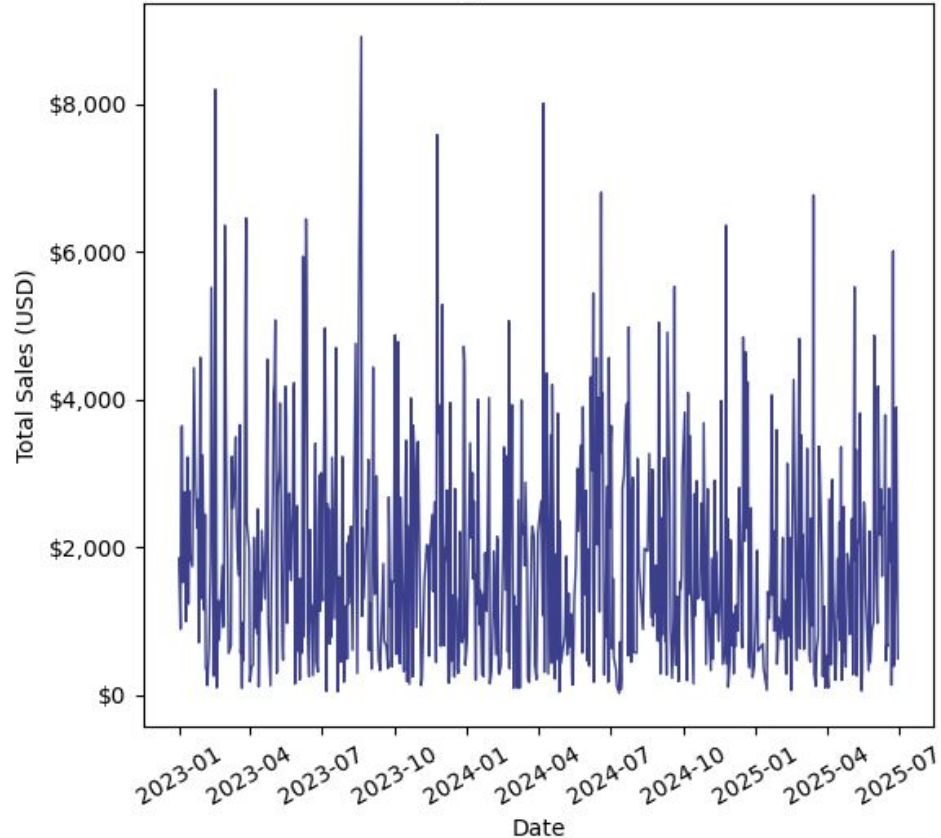


Sales by day

Daily revenue totals ordered chronologically – useful for trend and seasonality analysis.

E-Commerce Sales Analysis

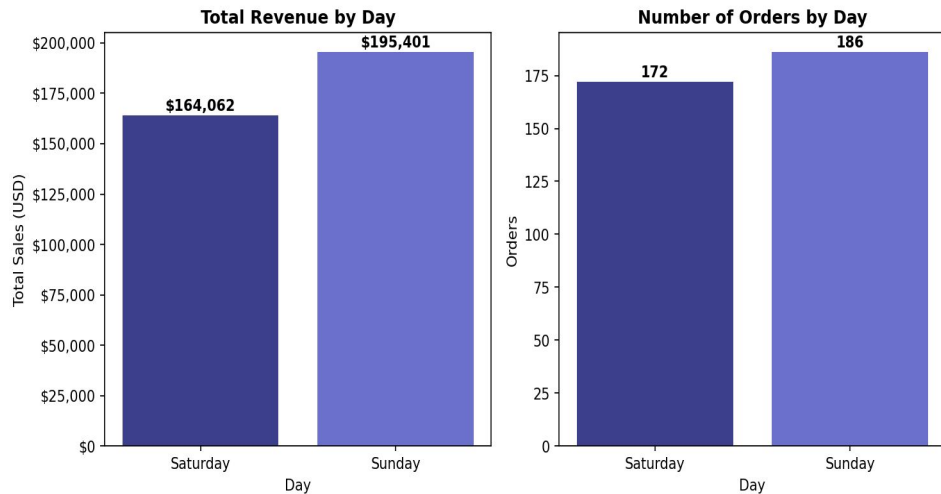
Daily Sales Trend



Weekend Sales Analysis

Total revenue on weekends only,
broken down by day name
(Saturday / Sunday), joined
through dim_fecha.

Weekend Sales Analysis





Thanks